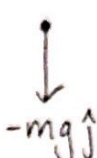
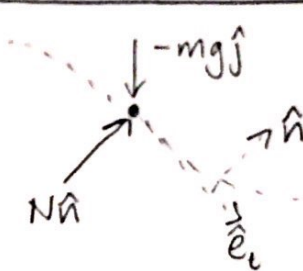


	Cart	FBD	Egns
Free		 or 	$\vec{F} = m\vec{a}$ $-mg\hat{j} = m(\ddot{x}\hat{i} + \ddot{y}\hat{j})$
Constrained	 Kinematic constraint (geometry of motion)		$N\hat{n} - mg\hat{j} = m(\ddot{x}\hat{i} + \ddot{y}\hat{j})$

adding constraint is different from adding forces

If try to solve directly, can't because we don't know $N(x, \dot{x}, y, \dot{y})$.

→ choices:

① Use minimal coordinates, say x , and pick N-E eqns that don't contain N

ex) $\{\vec{F} = m\vec{a}\} \cdot \vec{v} \Rightarrow$ one ODE in $x, \dot{x}, \ddot{x}$

② Use maximal coordinates

• Keep all coordinates, add constraint eqn $\Rightarrow$ 3 ODEs
How work in Matlab?

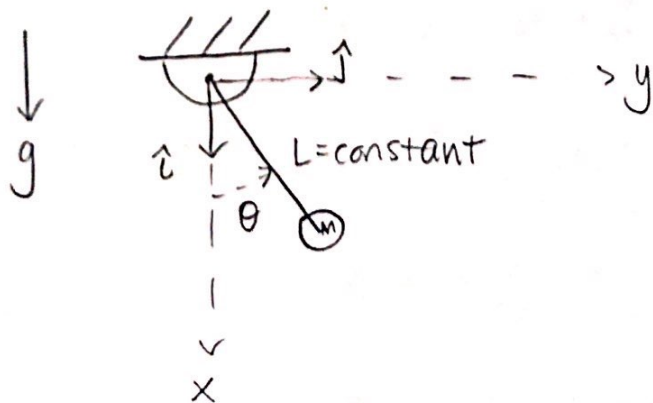
$$\begin{bmatrix} m & 0 & \square \\ 0 & m & \square \\ \square & \square & 0 \end{bmatrix} \begin{bmatrix} \ddot{x} \\ \ddot{y} \\ N \end{bmatrix} = \begin{bmatrix} \square \\ \square \\ \square \end{bmatrix}$$

$\square$: are terms w/ $x, \dot{x}, y, \dot{y}$ parameters

Matlab Implementation

```
zdot = myrhs(t,z,p)
% unpack to get x,y,vx,vy
a = [m 0 0;
      0 m 0;
      0 0 0];
b = [0 0 0]';
w = A\b
ax = w(1);
ay = w(2);
N = w(3);
```

Pendulum (Swingset)

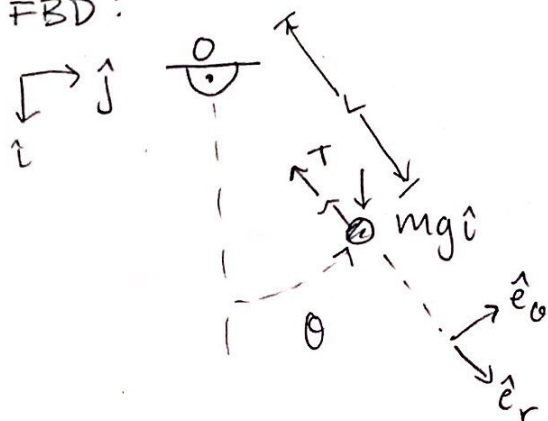


$$\text{EOM: } \ddot{\theta} = -\frac{g}{L} \sin \theta$$

→ How to find?

Method 1

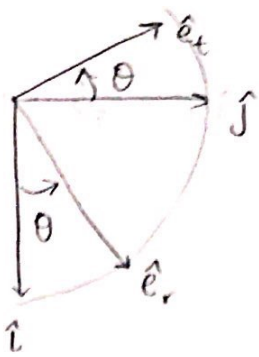
FBD:



AMB/O

$$\sum \vec{M}_{/O} = \dot{\vec{H}}_{/O}$$

$$\underbrace{\vec{r}_{/O}}_{L \hat{e}_r} \times (mg \hat{e}_y + -T \hat{e}_r) = m \underbrace{\vec{r}_{/O} \times \vec{a}}_{(\cancel{\dot{r}^0} - r\dot{\theta}^2) \hat{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta}) \hat{e}_\theta}$$



$$\hat{e}_r \times \hat{e}_\theta = \vec{0}$$

$$\hat{e}_\theta \times \hat{e}_r = \vec{0}$$

$$\hat{i} \times \hat{j} = \hat{e}_r \times \hat{e}_\theta = \hat{k}$$

$$\{ -\sin\theta mgl \hat{k} = m \hat{k} l \ddot{\theta} \} \cdot \hat{k}$$

$$\boxed{\ddot{\theta} = -\frac{g}{L} \sin\theta}$$

Method 2:

Same FBD

$$\left(\begin{array}{c} \vec{F} = m\vec{a} \\ \uparrow \quad \quad \uparrow \\ mg\hat{i} - T\hat{e}_r \end{array} \right) \cdot \hat{e}_\theta = -\dot{\theta}^2 l \hat{e}_r + l \ddot{\theta} \hat{e}_\theta$$

ex) Conservation of Energy

$$E_k = \frac{1}{2} m \underbrace{l^2 \dot{\theta}^2}_{v^2}$$

$$E_p = -l \cos\theta \cdot mg$$

$$E_T = \text{constant} = \frac{1}{2} m l^2 \dot{\theta}^2 - l \cos\theta \cdot mg$$

$$0 = \dot{E} = m l^2 \cancel{\dot{\theta}} \ddot{\theta} + l mg (\sin\theta) \cancel{\dot{\theta}}$$

→ Assume $\dot{\theta} \neq 0$

$$\boxed{\ddot{\theta} = -\frac{g}{L} \sin\theta}$$